Matrix Inverse Proof

Let

$$A = \begin{pmatrix} 3 & 1 & -1 \\ -1 & 2 & 0 \\ -1 & -1 & -1 \end{pmatrix}$$

Solve $A\vec{x} = \vec{b}$ where:

1)

$$\vec{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

2)

$$\vec{b} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}$$

Determinant of A:

$$\det(A) = 3(-2) - 1(-1) + (-1)(3)$$

$$\det(A) = -10$$

Minor matrix:

$$M = \begin{pmatrix} -2 & 1 & 3 \\ -2 & -4 & -2 \\ 2 & -1 & 7 \end{pmatrix}$$

Cofactor matrix:

$$C = \begin{pmatrix} -2 & -1 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 7 \end{pmatrix}$$

$$\text{Transpose} = \begin{pmatrix} -2 & 2 & 2 \\ -1 & -4 & 1 \\ 3 & 2 & 7 \end{pmatrix}$$

$$A^{-1} = \frac{1}{-10} \begin{pmatrix} -2 & 2 & 2 \\ -1 & -4 & 1 \\ 3 & 2 & 7 \end{pmatrix}$$

$$A\vec{x} = \vec{b}$$

$$A^{-1}A\vec{x} = A^{-1}\vec{b}$$

$$I\vec{x} = A^{-1}\vec{b}$$

$$\vec{x} = A^{-1}\vec{b}$$

$$1. \quad \vec{b} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$$

$$A^{-1}\vec{b} = -\frac{1}{10} \begin{pmatrix} -4 \\ -12 \\ -4 \end{pmatrix}$$

$$2. \quad \vec{b} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A^{-1}\vec{b} = A^{-1} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Matrices 2: Solving Square Systems of Linear Equations; Inverse Matrices

- Solving square systems of linear equations using inverse matrices.Solving square systems of linear equations; inverse matrices. Linear algebra is essentially about solving systems of linear equations, an important application of mathematics to real-world problems in engineering, business, and science, especially the social sciences. Here we will just stick to the most important case, where the system is square, i.e., there are as many variables as there are equations. In low dimensions]

Linear algebra is essential. Such systems look as follows:

(We are given a 2×2 system and a 3×3 system.)

$$7. \quad \begin{cases} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{cases}$$

$$\begin{pmatrix} a_{11}x_1 & a_{12}x_2 & a_{13}x_3 \\ a_{21}x_1 & a_{22}x_2 & a_{23}x_3 \\ a_{31}x_1 & a_{32}x_2 & a_{33}x_3 \end{pmatrix}$$

In this system, the a_{ij} and b_i are given, and we solve for the x_i.

As a simple mathematical example, consider the linear change of coordinates given by the equation

$$\begin{aligned} x_1 &= a_{11}y_1 + a_{12}y_2 + a_{13}y_3, \\ x_2 &= a_{21}y_1 + a_{22}y_2 + a_{23}y_3, \\ x_3 &= a_{31}y_1 + a_{32}y_2 + a_{33}y_3. \end{aligned}$$

We know the y-coordinates of a point; then these equations tell its x-coordinates immediately. But if instead we are given the x-coordinates, we must find the y-coordinates and solve equation (7) above using matrix multiplication.

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$$Ax = b, \qquad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Where A is the square matrix of coefficients (a_{ij}) of the 2×2 system, and the $n \times n$ system would be written analogously. All of them are abbreviated by the same equation $Ax = b$.

You have had experience with solving small systems like (7) by elimination: multiplying the equations by constants and subtracting them from each other, the purpose being to eliminate all the variables but one. When elimination is done systematically, it is an efficient method. Here however we want to talk about another method more compatible with hand held calculators and MatLab, and which leads more rapidly to certain key ideas and results in linear algebra.

Inverse Matrices

Referring matrices to the system (8), we can find a square matrix M, the same size as A, such that

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$$MA = I$$

(where I is the identity matrix)

Matrix Inverses

We can then solve (8) by matrix multiplication, using the successive steps

$$Ax = b$$

$$M(Ax) = Mb$$

$$x = Mb$$

Where the step $M(Ax) = x$ is justified by

$$M(Ax) = (MA)x$$

$$= Ix$$

$$= x$$

by the associative law, because I is the identity matrix.

Example 2.7
Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}.$$

Verify that M satisfies (9) above, and use it to solve the first system below for x_i and the second for the y_i in terms of the x_i.

$$\begin{cases} x_1 + 2x_2 = -1 \\ 2x_1 + 3x_2 = 4 \end{cases} \qquad \begin{cases} x_1 = y_1 + 2y_2 \\ x_2 = 2y_1 + 3y_2 \end{cases}$$

Solution: we have

$$\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

By matrix multiplication, for the first system we have

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 11 \\ -6 \end{pmatrix}$$

So the solution is

$$x_1 = 11, \qquad x_2 = -6.$$

By reasoning similar to that used above in going from $Ax = b$ to $x = Mb$, the solution to $x = Ay$ is $y = Mx$, so we get

$$y_1 = -3x_1 + 2x_2$$

$$y_2 = 2x_1 - x_2$$

Our problem now is:

$$M \text{ exists} \iff |A| \neq 0$$

The implication follows immediately from the law

$$\det(AB) = \det(A)\det(B),$$

since

$$MA = I \Rightarrow |M||A| = |I| = 1 \Rightarrow |A| \neq 0.$$

Matrix Inverses

The implication in the direction requires more; for the low-dimensional cases, we will provide a formal definition first and give M its proper name A^{-1}.

Definition

Let A be an $n \times n$ matrix with $|A| \neq 0$. Then the inverse of A is an $n \times n$ matrix, written A^{-1}, such that

$$A^{-1}A = I_n \quad \text{and} \quad AA^{-1} = I_n.$$

It is actually enough to verify either equation; the other follows automatically.

Consequences

If $|A| \neq 0$, then the unique solution of

$$AX = b$$

is

$$X = A^{-1}b.$$

Calculating the Inverse of a 3×3 Matrix

$$A^{-1} = \frac{1}{|A|}\,\mathrm{Adj}(A) = \frac{1}{|A|}\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}.$$